

# Parallel Document Similarity Search

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# | MOTIVATION

## Explosive Growth

The volume of digital text data is growing exponentially. Traditional sequential methods cannot scale to petabyte level requirements without massive delays.

## Critical Need

Applications like plagiarism detection and search engines require sub-second response times. High performance parallelism is the only viable path.

Goal: Optimize similarity search through distributed and shared memory parallelism.

# PROBLEM STATEMENT



## Top-K

RESULT EXTRACTION

### The Objective

Input:  $n$  documents and a query vector.

Output: The  $K$  most similar documents identified by minimizing the total execution time.

$$T = T_{IO} + T_{scatter} + T_{compute} + T_{Gather}$$

# KEY CHALLENGES



## Complexity

$O(n \times m)$  compute cost for high-dimensional vectors.



## Sorting

Inefficient  $O(n \log n)$  sorting of global result sets.



## Latency

High MPI communication overhead between cluster nodes.



## Imbalance

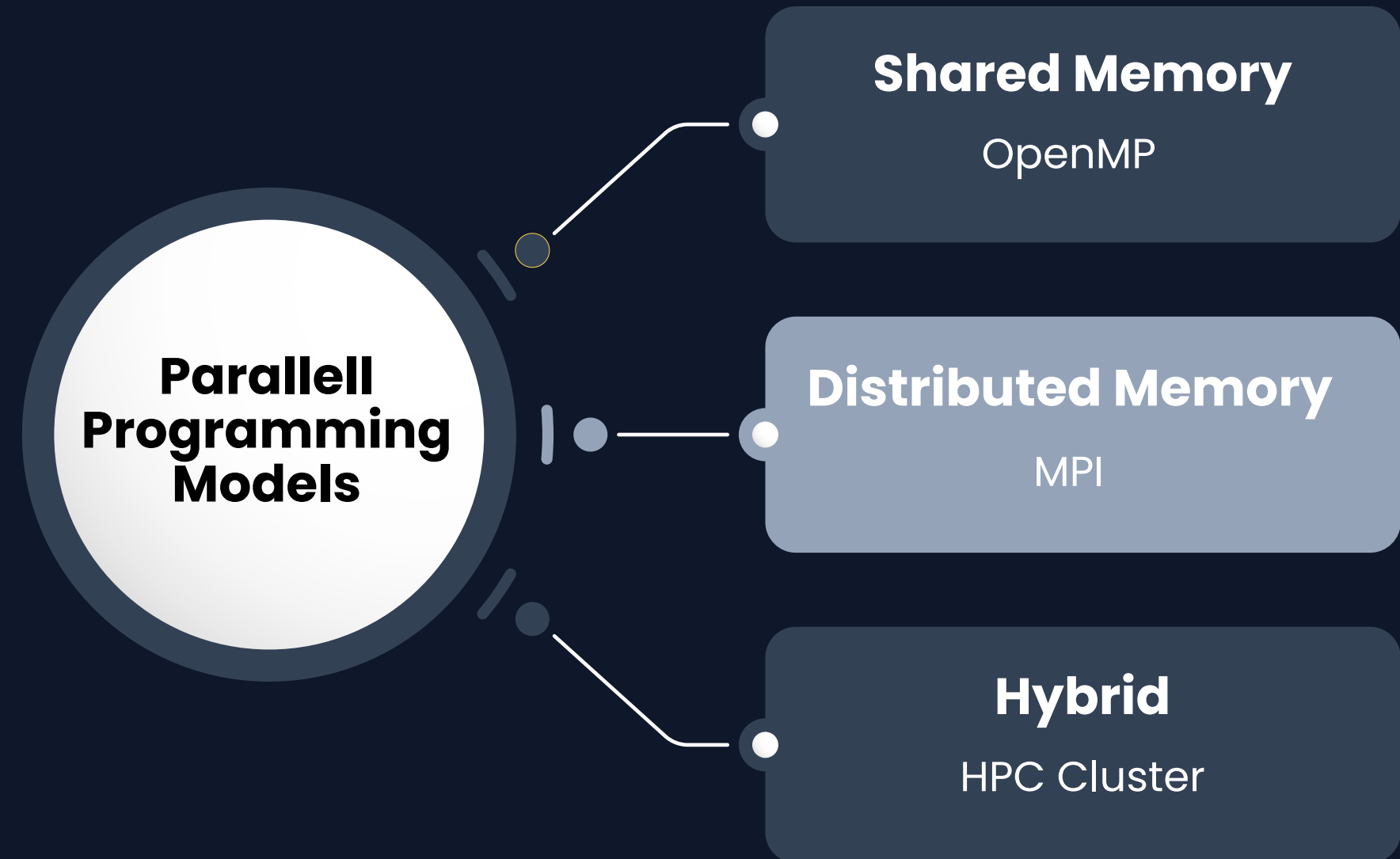
Uneven workload distribution across CPU cores.

# | PROPOSED APPROACH

 **MPI:** Inter-node parallelism for distributed cluster systems.

 **OpenMP:** Intra-node multi-threading for local shared memory.

 **Min-Heap:** Optimized Top-K selection to reduce memory footprint.





# System Architecture

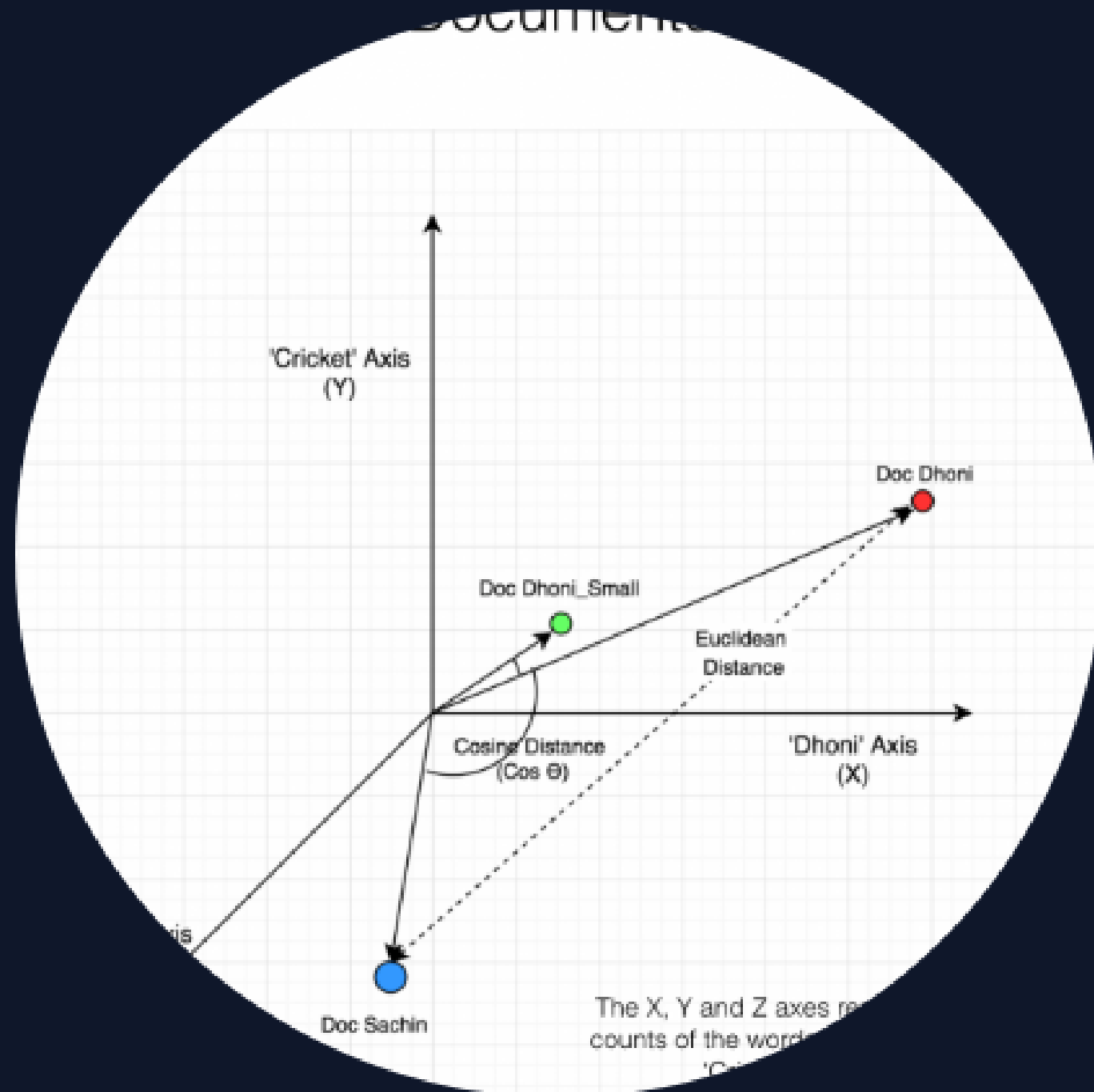
## | ALGORITHM & OPTIMIZATION



🛡️ Optimization Note: Sentinel padding used to handle non-uniform document chunks.

Two-Phase Design: Phase A (OpenMP parallel) — compute scores into shared array. Phase B (sequential) — build heap. Avoids race conditions on heap.

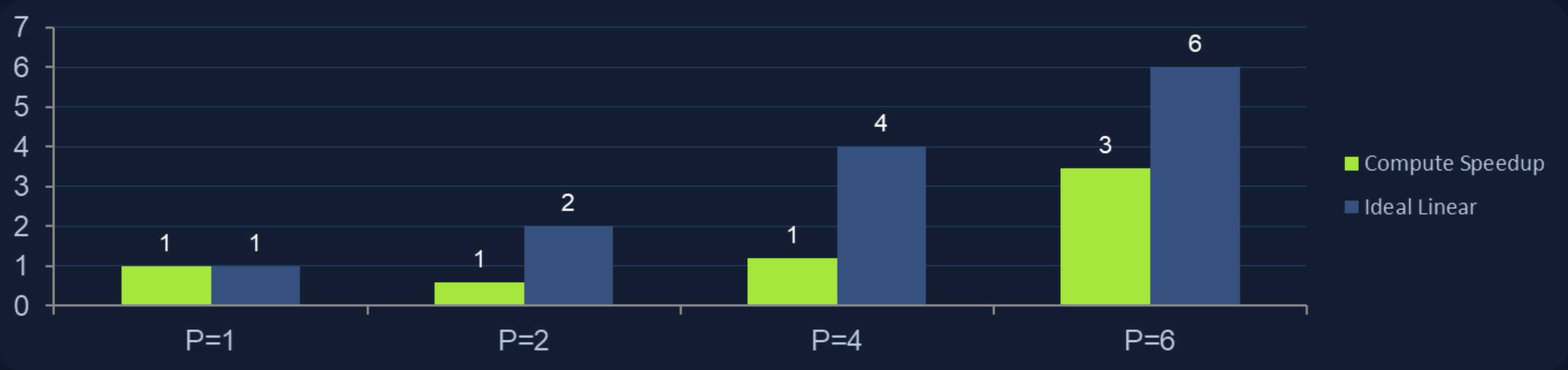
# SIMILARITY METRICS



- ^ **Cosine Similarity:** Measures orientation rather than magnitude.
- ⌚ **Euclidean Distance:** Straight-line L2 norm measurement.
- A **Manhattan Distance:** Absolute coordinate sum (L1 Norm).
- ⚗ **Power Similarity:** Parameterized non-linear relation detection.

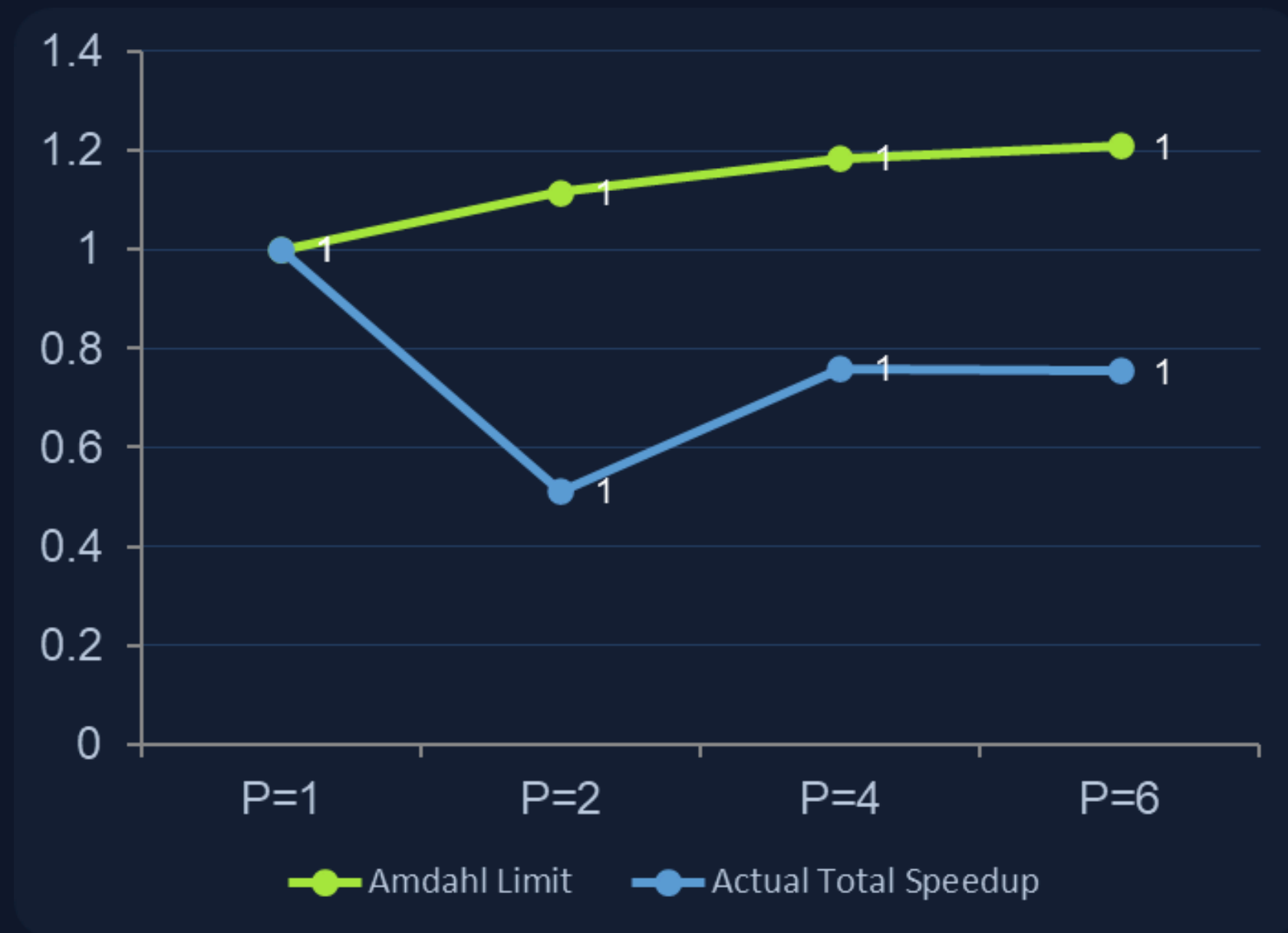
# RESULTS & SPEEDUP

| MPI Ranks | I/O (s)  | Scatter (s) | Compute (s) | Total (s) | Speedup (computation) |
|-----------|----------|-------------|-------------|-----------|-----------------------|
| P = 1     | 0.009724 | 0.000375    | 0.002113    | 0.012282  | 1.00× (baseline)      |
| P = 2     | 0.016951 | 0.001948    | 0.003627    | 0.023967  | 0.58× (WSL2 artefact) |
| P = 4     | 0.01276  | 0.001506    | 0.001771    | 0.016153  | 1.19×                 |
| P = 6     | 0.013809 | 0.001273    | 0.000613    | 0.016265  | 3.45×                 |



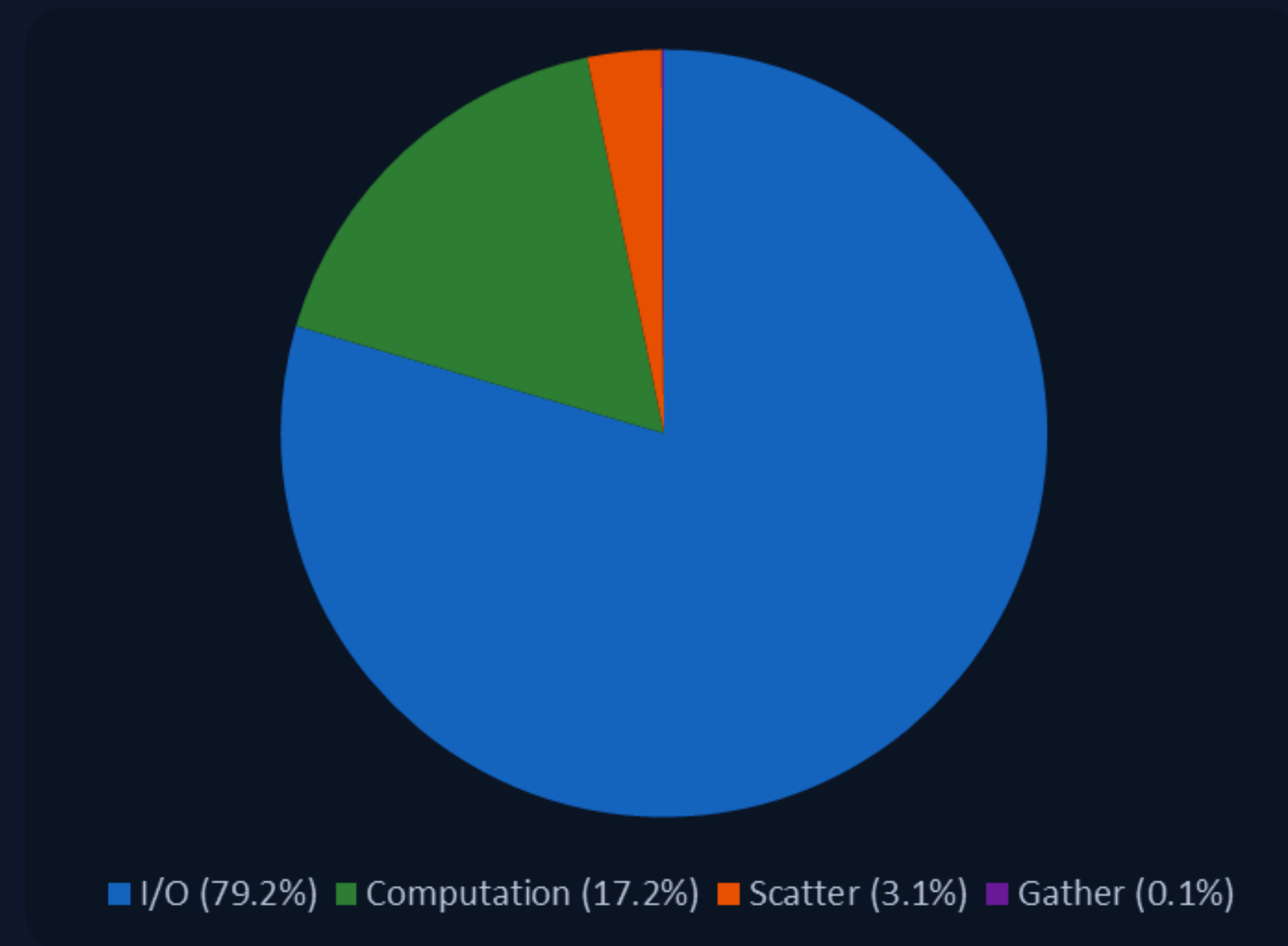


# PERFORMANCE ANALYSIS



**Amdahl's Law:  $f = 0.792 \rightarrow$  Max total speedup =  $1.26\times$**

Actual speedup stays below Amdahl bound because Amdahl assumes zero communication cost.



I/O = 79.2% of runtime. Hard ceiling:  $S(\infty) = 1/0.792 \approx 1.26\times$

# CONCLUSION & FUTURE

- ✓ **Efficient System:** Achieved a scalable hybrid framework for document search.
- 🧪 **Scalability:** Successfully handled distributed datasets with high precision.
- 🔑 **GPU Search:** Future roadmap includes CUDA acceleration.
- 💾 **MPI-IO:** Improving ingestion with parallel disk access.

Thank You!

